

RITIKA RAJESH KUMAR JHA

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EDUCATION

B.M.S College of Engineering

B. E in Electronics and Telecommunication

Minors in Data Science

CGPA - **8.915** | *Ranked 4th in a class of 70 students*

Relevant Coursework: Digital electronics, Digital Signal Processing, Analog Circuits, VLSI Design, Wireless Communication Systems, Control Systems, Probability & Statistics, Verilog HDL.

HONORS, AWARDS & GRANTS

- **IEI Karnataka SRPP Grant Recipient** – Awarded a ₹3,000 research grant for developing a 10.525 GHz antenna system for AI-based activity monitoring.
- **CICAD 2026–27 Merit Scholar** – Selected through a merit-based process for the Comprehensive Internship Course on Antenna Design conducted by RaoS Consultants LLC, Sanyark Space Technologies, and the WAMS Society.
- **ISWDP Cohort 7 Scholar** – Received a 75% merit scholarship for academic excellence and advanced training in semiconductors and chip design using Synopsys TCAD.
- **MedNxt Hackathon Winner (1st Place)** – Developed an AI-based healthcare solution and secured first place among competing teams.

EXPERIENCE

U R Rao Satellite Centre (ISRO)

June 2026 – Aug 2026

Research Intern – Antenna & Passive Design Systems (APSD)

- Designed and simulated microwave passive components, including microstrip T-junctions, 1:2 and 1:3 Wilkinson Power Dividers, using circuit-level synthesis and full-wave electromagnetic simulation in ANSYS HFSS.
- Translated transmission-line circuit designs into microstrip structures and evaluated impedance matching, power division, return loss, insertion loss and port isolation using S-parameter analysis.
- Analysed electromagnetic performance around 8.125 GHz, developing practical experience in microwave component design and EM simulation for space-oriented systems.

Heligo Charters Pvt. Ltd.

June 2025 – July 2025

Avionics Intern

- Analysed avionics telemetry and sensor datasets using time- and frequency-domain techniques.
- Investigated signal integrity, subsystem interactions, and fault patterns across aircraft systems.
- Performed anomaly analysis on HUMS/QAR data to support reliability assessment and troubleshooting.

PROJECTS

Design and Optimization of a 10.525 GHz X-Band Microstrip Antenna Array for DFHAR

Ongoing Major Project

- Designed and validated four microstrip antenna configurations: 2.4 GHz single patch, 10.525 GHz single patch, 2×2 array at 2.4 GHz, and 2×2 array at 10.525 GHz using ANSYS HFSS.
- Optimized antenna performance through impedance matching, radiation pattern analysis, gain enhancement, and bandwidth evaluation.

- Developing the antenna system for 10.525 GHz Doppler radar-based human activity monitoring, integrating RF sensing with signal-processing techniques.

Condition Monitoring and Fault Detection Using Sensor Data

Project work

- Developed an ESP32-based condition monitoring system using Arduino IDE, integrating temperature, vibration, and accelerometer sensors for real-time data acquisition.
- Implemented embedded C firmware for sensor interfacing, Bluetooth-enabled communication, and continuous system monitoring.
- Applied signal processing and anomaly detection techniques to identify abnormal operating conditions and assess system reliability.

Embedded Hardware Design Portfolio using KiCad

Ongoing Independent Project

- Designing a portfolio of embedded hardware PCBs in KiCad, including regulated power supplies, solar-powered systems, and an MCU-based EEPROM datalogger with RTC support.
- Developing complete schematics and PCB layouts while building expertise in power management, embedded hardware design, and PCB layout optimization.
- Applying PCB design principles including component selection, power distribution, routing, connector placement, design rule checks (DRC), and preparation for fabrication.

CERTIFICATIONS

- **Embedded Systems and Sensor Interfacing** (Udemy)
- **MATLAB Certifications** (MathWorks)

LEADERSHIP

- Student Placement Coordinator, BMSCE
- ETE Coding Club Coordinator, UTSAV 2025

TECHNICAL SKILLS

- **RF, Microwave & Analog:** Microstrip Antenna Design, Microwave Passive Components, Transmission Line Theory, Impedance Matching, S-Parameter Analysis, ANSYS HFSS.
- **Signal Processing:** Digital Signal Processing, FFT, FIR Filters, Spectral Analysis, Noise Reduction, Time-Series Analysis, Wireless Communication Systems
- **EDA and Simulation Tools:** KiCAD (Learning), Cadence Virtuoso (Schematic Capture & Basic Analog Circuit Simulation), LTspice, Multisim, Ansys HFSS, Snopsys TCAD, Proteus
- **Hardware:** Digital Electronics, Verilog HDL, VLSI Design, FSM Design, Counters & Registers, Timing Analysis, CMOS Fundamentals, Digital System Design
- **Programming:** Python, MATLAB
- **Embedded Systems:** Embedded C, ESP32, Microcontrollers, Sensor Interfacing, Interrupts, GPIO